

Movie Success Prediction

OSUPR

Luka Mišmaš

1. Introduction

Predicting the commercial success of a movie before its release is a challenging but valuable problem. Film studios invest significant resources into production, and the ability to estimate whether a movie will be profitable can inform both financial and creative decisions.

This project uses machine learning to predict whether a movie will be commercially successful based on pre-release information such as budget, runtime, genre, and cast/crew data. Three classification models are compared: Logistic Regression, Random Forest, and an Artificial Neural Network (ANN).

2. Data

2.1 Dataset

The dataset used is the TMDB 5000 Movie Dataset, which contains information on approximately 5,000 movies. It consists of two CSV files: `tmdb_5000_movies.csv` and `tmdb_5000_credits.csv`, which were merged on the movie ID.

After merging and cleaning, the final dataset contained approximately 3,800 movies with non-zero budget and revenue values.

2.2 Success Label

A movie was labeled as successful (1) if its revenue was at least 1.5 times its budget. This threshold was chosen to reflect a reasonable return on investment after accounting for marketing and distribution costs. Movies below this threshold were labeled as unsuccessful (0).

The resulting class distribution was approximately balanced, which ensured reliable model training.

2.3 Features

The following features were used as model inputs:

- Budget
- Runtime
- Popularity score
- Average vote count
- Vote count
- Release month
- Director experience — number of movies the director has in the dataset
- Cast fame score — average appearance frequency of the top 3 cast members across the dataset
- Binary genre indicators for the 10 most common genres

Director experience and cast fame were custom-engineered features added to improve model performance. They are based on the hypothesis that more experienced or recognizable talent correlates with higher success rates.

3. Methods

3.1 Preprocessing

Movies with missing or zero budget/revenue values were removed. Missing runtime and release date values were dropped. The release date was converted to a release month feature. Missing director experience values were filled with 1, and missing cast fame scores were filled with 0.

All numeric features were standardized using scikit-learn's StandardScaler before training. The dataset was split into 80% training and 20% test sets using a fixed random seed for reproducibility.

3.2 Models

Three classification models were trained and evaluated:

- Logistic Regression — a linear baseline model with a maximum of 1000 iterations.
- Random Forest — an ensemble of 100 decision trees.
- ANN (MLP) — a multi-layer perceptron with hidden layers of sizes (128, 64, 32) and a maximum of 1000 iterations.

Each model was evaluated using accuracy, precision, recall, and F1 score on the held-out test set.

4. Results

4.1 Model Performance

The table below summarizes the performance of all three models on the test set:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.738	0.784	0.823	0.803
Random Forest	0.772	0.794	0.876	0.833
ANN (MLP)	0.745	0.777	0.850	0.812

Random Forest achieved the best overall performance with an F1 score of 0.833. Logistic Regression and ANN also performed reasonably well, with F1 scores of 0.803 and 0.812 respectively.

All three models show higher recall than precision, meaning they tend to predict more movies as successful than are actually successful. This is likely due to a slight class imbalance in the dataset favoring successful films.

4.2 Feature Importance

Based on the Logistic Regression coefficients, the most influential features for predicting success were budget, popularity, and vote count. Genre features had a smaller but non-

negligible contribution. Director experience and cast fame showed a moderate positive association with success.

5. Discussion and Conclusion

The Random Forest model performed best across all metrics and is the recommended model for this task. Its ability to capture non-linear relationships between features likely explains its advantage over Logistic Regression.

The ANN underperformed relative to expectations, which is likely due to the relatively small dataset size. Neural networks typically require more data to generalize well, and the 3,800-sample dataset may not be sufficient to fully exploit the model's capacity.

A key limitation of this study is that all features are measured post-release (e.g., popularity and vote data may reflect post-release trends). In a real-world deployment, only pre-release features such as budget, cast, director, and genre would be available.

Future work could include more advanced feature engineering, hyperparameter tuning, and the use of larger or more recent datasets to improve generalization.

References

TMDB 5000 Movie Dataset. Retrieved from Kaggle: <https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>

Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.